Generalised Eccentricity, Radius and Diameter in Graphs

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Abstract

For a vertex v and a $(k-1)$-element subset P of vertices of a graph, one can define the *distance* from v to P in various ways, including the minimum, average, and maximum distance from v to P. Associated with each of these distances, one can define the k-eccentricity of the vertex v as the maximum distance over all P, and the k-eccentricity of the set P as the maximum distance over all v. If $k = 2$ one is back with the normal eccentricity. We study here the properties of these eccentricity measures, especially bounds on the associated radius (minimum k-eccentricity) and diameter (maximum k-eccentricity).

1 Introduction

The concept of eccentricity is fundamental in graph theory. If $G = (V, E)$ is a connected graph, the *distance* $d(u, v)$ between two vertices u and v of G is defined as the minimum length of a u–v path of G. The *eccentricity* $e(v)$ of v is $\max_{u\in V} d(u, v)$. That is, $e(v)$ is the distance between v and a vertex farthest from v. The radius *rad* G of G is the minimum eccentricity among the vertices of G, and the diameter *diam* G of G is the maximum eccentricity. It is well known that the radius and diameter are related by the inequality $rad\ G \le diam\ G \le 2\, rad\ G$.

Several authors have considered the question of how to define the distance between two sets of vertices (see [1]). If A and B are not necessarily disjoint sets of vertices, we define

- the **minimum distance** from A to B as

$$d_{min}(A, B) = \min\{\, d(a, b) : a \in A,\ b \in B \,\},$$

- the **average distance** from A to B as

$$d_{ave}(A, B) = \frac{1}{|A||B|} \sum_{a\in A,\, b\in B} d(a, b), \text{ and}$$

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- the **maximum distance** from A to B as

$$d_{max}(A, B) = \max\{\, d(a, b) : a \in A,\ b \in B \,\}.$$

If either set contains only one vertex we will identify the vertex with the set. Other ways of defining the distance between two sets are possible, for example the **Steiner distance** (see [2, 3]).

The normal eccentricity is based on the distance between two vertices. Consider the generalisation to more vertices. There are at least two possible ways to proceed.

- The eccentricity is defined for a single vertex v as the maximum distance between v (or if you prefer $\{v\}$) and a set of $k-1$ vertices. That is, for a given distance measure d_M the **k-eccentricity of a vertex** is:

  $$e_M(v, k) = \max\{\, d_M(v, P) : P \subseteq V,\ |P| = k - 1 \,\}.$$

  If $k = 2$, then all the above definitions of distance yield the normal definition of eccentricity.

- The eccentricity is defined for a set P of $k-1$ vertices as the maximum distance between P and another vertex. That is, **the eccentricity of a set** is:

  $$e_M(P) = \max\{\, d_M(P, v) : v \in V \,\},$$

  If $|P| = 1$, then all the above definitions of distance yield the normal definition of eccentricity.

The **k-radius** and the **k-diameter** of the graph are then defined as the minimum and maximum k-eccentricities among all vertices in the graph. The **k-set-radius** and the **k-set-diameter** of the graph are the minimum and maximum eccentricities among all $(k-1)$-element subsets of the vertices. Of course, the k-diameter and the k-set-diameter are equal. We will assume throughout that k is at least 2 and at most the order of the graph.

For the first three definitions of distance, the k-eccentricity of a vertex is intimately related to its eccentricity. For example,

$$e(v) - k + 1 \le e_{min}(v, k) \le e_{ave}(v, k) \le e_{max}(v, k) = e(v).$$

Thus the k-radius and k-diameter parameters behave much like the normal radius and diameter (at least for orders much larger than k), and are in this sense uninteresting to study. (But see [5, 4].)

Moreover, if one considers the eccentricity of a set, say P, then

$$e_{max}(P) = \max\{\, e(w) : w \in P \,\}.$$

As a result the k-set-radius and k-set-diameter for maximum distance behave much like the normal radius and diameter, and we do not study them here.

The parameters we do study are the k-set-eccentricity for the distance measure d_{min} (Section 2), and the k-set-eccentricity for the distance measure d_{ave} (Section 3). In particular, we establish results connecting the new and normal radius and diameter.

The optimality of several results is shown by the following class of trees. For positive integers a, b, c, define $T(a, b, c)$ as the tree obtained by taking $a-1$ paths of length b and one path of length c and identifying an endpoint of each path. Equivalently, take the star with a edges, then subdivide one edge $c-1$ times and the remainder $b-1$ times. For example, $T(2, b, c)$ is a path, $T(a, 1, 1)$ is a star, $T(a, b, b)$ is sometimes called an octopus, and $T(a, 1, c)$ is sometimes called a comet or broom. The tree $T(a, b, c)$ has order $(a-1)b + c + 1$.

2 The Min-Set-Eccentricity

In this section we consider the k-set-eccentricity resulting from defining the distance between two sets as the minimum distance between a vertex from each.

For $k \geq 2$, we denote the min-k-set-radius of G as

$$rad_{min}(G, k) = \min\{\, e_{min}(P) : P \subseteq V,\ |P| = k-1 \,\},$$

and the min-k-set-diameter of G as

$$diam_{min}(G, k) = \max\{\, e_{min}(P) : P \subseteq V,\ |P| = k-1 \,\}.$$

If S is a set of $(k-1)$ vertices for which $e_{min}(S)$ equals $rad_{min}(G, k)$, then S is called a $(k-1)$-*center* of G. Thus a t-center is a set S of t vertices such that every vertex is close to S, i.e., to at least one vertex of S. Applications of t-centers in graphs occur in facility location problems. For example, if a city wishes to locate t fire-stations in such a way as to minimise the response time to the farthest point being protected, then this is a t-center problem. The t-center problem is discussed in the book by Buckley and Harary [1].

We begin with the following lemma which shows that a k-center cannot be too spread out.

Lemma 1 *Let G be a connected graph, and let $k \geq 3$ be an integer. Let C be a $(k-1)$-center of G. If C' and C'' form a partition of C, then $d_{min}(C', C'') \leq 2\, rad_{min}(G, k) + 1$.*

Proof. Suppose $d_{min}(C', C'') = d(u, v) = \ell$, where $u \in C'$ and $v \in C''$. Let P be a u–v path of length ℓ, and let w be a central vertex of P. We may assume that $d(u, w) = \lfloor \ell/2 \rfloor$ and $d(v, w) = \lceil \ell/2 \rceil$. Then $rad_{min}(G, k) = e_{min}(C) \geq d_{min}(w, C) = d(u, w) = \lfloor \ell/2 \rfloor$. Thus, $d(C', C'') = \ell \leq 2\, rad_{min}(G, k) + 1$. □

We can now relate the ordinary radius and the k-set-radius of a graph.

Theorem 2 *Let G be a connected graph, and let $k \geq 2$ be an integer. Then*

$$rad_{min}(G, k) \leq rad\ G \leq (k-1)\, rad_{min}(G, k) + \left\lceil \frac{k-2}{2} \right\rceil,$$

and these bounds are sharp.

Proof. To prove the lower bound, let v be a central vertex of G and let S be any $(k-1)$-element subset of V containing v. Then $d_{min}(u,S) \le d(u,v) \le e(v) = rad\ G$ for every vertex u, and so $rad_{min}(G,k) \le e_{min}(S) \le rad\ G$. This establishes the lower bound.

That the lower bound is sharp may be seen by considering the octopus $T(k,r,r)$ of radius r. For any $(k-1)$-element subset S of $T(k,r,r)$, at least one of the branches contains no vertex of S, so $e_{min}(S) \ge r$. Therefore $rad_{min}(T(k,r,r)) = r$.

Next we prove the upper bound. Set $m = rad_{min}(G,k)$. Let C be a $(k-1)$-center of G. We construct a sequence of trees T_1, T_2, $\cdots$, T_{k-1} inductively as follows. Let c_1 be any vertex in C, and let T_1 be the subtree consisting of the vertex c_1. Then for $i = 1, 2, \ldots, k-2$, let c_{i+1} be the vertex of $C_i = C - \{c_1, \ldots, c_i\}$ closest to T_i; that is, $d_{min}(c_{i+1}, V(T_i)) = d_{min}(C_i, V(T_i))$. By Lemma 1, $d_{min}(C_i, V(T_i)) \le d_{min}(C_i, \{c_1, \ldots, c_i\}) \le 2m+1$. Let T_{i+1} be the tree obtained from T_i by adding a shortest path from c_{i+1} to T_i. The tree T_{k-1} so constructed contains all the vertices of C and has diameter at most $(k-2)(2m+1)$. Let w be a central vertex of T_{k-1}. Then any vertex of G is within distance m from C, and any vertex of C is within distance $rad(T_{k-1})$ from w. Hence $e_G(w) \le rad(T_{k-1}) + m \le (k-1)m + \lceil (k-2)/2 \rceil$. This establishes the upper bound.

That the upper bound is sharp may be seen as follows. Let G be the path $u_1, u_2, \ldots, u_n$ on $n = 2\ell(k-1) + k - 1$ vertices. Let S be the $(k-1)$-element subset of vertices of G given by $S = \{u_{\ell+1},\ u_{3\ell+2},\ u_{5\ell+3},\ \ldots,\ u_{(2k-3)\ell+k-1}\}$. Every vertex of G is within distance ℓ from a unique vertex of S, while u_1, for example, is at distance exactly ℓ from S, so $e_{min}(S) = \ell$. Hence $rad_{min}(G,k) = e_{min}(S) = \ell$. Thus, $rad\ G = \lceil (n-1)/2 \rceil = \ell(k-1) + \lceil (k-2)/2 \rceil = (k-1)\, rad_{min}(G,k) + \lceil (k-2)/2 \rceil$. □

An immediate consequence of the proof of the upper bound of Theorem 2 now follows.

Corollary 3 *If C is a $(k-1)$-center of a connected graph G, where $k \ge 3$, then the distance between two vertices of C is at most $(k-2)(2\, rad_{min}(G,k) + 1)$.*

The following result establishes a sharp upper bound on the k-set-radius of a graph in terms of its order.

Theorem 4 *Let G be a connected graph of order n, and let $k \ge 2$ be an integer. Then*

$$rad_{min}(G,k) \le \frac{n}{k},$$

and this bound is sharp.

Proof. We proceed by induction on $k \ge 2$. If $k = 2$, then $rad_{min}(G,k) = rad\ G$. However, it is well-known that $rad\ G \le n/2$. Assume, then, that $rad_{min}(G',k') \le n/k'$ for all graphs G' and for all integers k' satisfying $2 \le k' < k$, where $k \ge 3$. We show that $rad_{min}(G,k) \le n/k$.

It is sufficient to prove the result for trees T. We consider two possibilities.

Case 1. There is an edge e such that both components of $T - e$ have order more than n/k.

Let $n_1 \le n_2$ be the orders of the components T_1 and T_2 of $T-e$. Then let $k_1 = \lceil n_1 k/n \rceil$ and $k_2 = k - k_1 + 1$. Then $2 \le k_1 \le k-1$ and $2 \le k_2 \le k-1$. Hence, applying the inductive hypothesis, we obtain $rad_{min}(T_i, k_i) \le n_i/k_i$ for $i = 1, 2$. Thus, for $i = 1, 2$, we can find $(k_i - 1)$-element subsets S_i of T_i such that in T_i, $d_{min}(v, S_i) \le n_i/k_i$ for all vertices v of T_i. However, by our choice of k_1 and k_2, $n_1/k_1 \le n/k$, and

$$\frac{n_2}{k_2} = \frac{n - n_1}{k - k_1 + 1} \le \frac{n - n_1}{k - \frac{n_1 k}{n}} = \frac{n}{k}.$$

Hence, $S = S_1 \cup S_2$ is a $(k-1)$-element subset S of T, such that for every vertex v of T, $d_{min}(v, S) \le n/k$; and so $rad_{min}(G, k) \le n/k$.

Case 2. For all edges e one component of $T - e$ has order at most n/k.

Let c be a centroid vertex of T. That is, for every edge e incident with c, the vertex c is in the larger component of $T - e$. However, by assumption, the smaller component of $T - e$ has at most n/k vertices. Hence every vertex of T is within distance n/k from c. Thus, $rad_{min}(T, k) \le rad\, T \le n/k$. This completes the proof of the upper bound.

That this bound is sharp may be seen by taking the subdivided star $T(k, r, r-1)$ of order $n = kr$. Then $rad_{min}(T(k, r, r-1), k) = r = n/k$. □

We now consider the k-set-diameter of a graph. The ordinary diameter and the k-set-diameter of a graph are related as follows.

Theorem 5 *Let G be a connected graph, and let $k \ge 2$ be an integer. Then*

$$diam\ G - k + 2 \le diam_{min}(G, k) \le diam\ G,$$

and these bounds are sharp.

Proof. The upper bound is immediate from the definitions of the parameters. To prove the lower bound, take a diametrical path in G, say a u–v path, so $d(u, v) = diam\ G$. Let S be the set consisting of the first $k-1$ vertices on this path. Then $diam_{min}(G, k) \ge e_{min}(S) \ge d_{min}(v, S) = diam\ G - k + 2$. This establishes the lower bound.

To see that the lower bound is sharp consider a path. To see that the upper bound is sharp consider the octopus $T(k, r, r)$. □

It is straight-forward to see that:

Theorem 6 *Let G be a connected graph of order n, and let $k \ge 2$ be an integer. Then*

$$diam_{min}(G, k) \le n - k + 1,$$

and this bound is sharp.

The k-radius and k-diameter of a graph are related by the following inequality.

Theorem 7 *Let G be a connected graph, and let $k \geq 2$ be an integer. Then*

$$diam_{min}(G,k) \leq 2(k-1)\, rad_{min}(G,k) + k - 2,$$

and this bound is sharp.

Proof. If $k=2$, then $diam\ G = diam_{min}(G,k)$ and $rad\ G = rad_{min}(G,k)$ and the bound is known. Assume, then, that $k \geq 3$ and set $m = rad_{min}(G,k)$. Let C be a $(k-1)$-center of G. Let u and v be two vertices of G with $d(u,v) = diam\ G$. Then each of u and v is within distance m from C, while any two vertices of C are at distance at most $(k-2)(2m+1)$ apart by Corollary 3. Hence $d(u,v) \leq 2m + (k-2)(2m+1) = 2(k-1)m + k - 2$. However, by Theorem 5, $diam_{min}(G,k) \leq diam\ G = d(u,v)$. The proof of the upper bound now follows.

That this inequality is sharp may be seen as follows. For integer $\ell \geq 2$, let G denote the comet $T(k, 1, (2\ell+1)(k-1)-2)$. Then $diam_{min}(G,k)$ is achieved by taking S to be the $k-1$ end-vertices incident with the vertex of maximum degree. On the other hand, let $u_1, u_2, \ldots, u_{(2\ell+1)(k-1)-1}$ denote the tail of the comet. Let S be the $(k-1)$-element subset of vertices of G given by $S = \{u_{\ell+1},\ u_{3\ell+2},\ u_{5\ell+3},\ \ldots,\ u_{(2k-3)\ell+k-1}\}$. Every vertex of G is within distance ℓ from some vertex of S, while u_1, for example, is at distance exactly ℓ from S, so $e_{min}(S) = \ell$. It follows that $rad_{min}(G,k) = e_{min}(S) = \ell$. Consequently, $diam_{min}(G,k) = 2(k-1)\, rad_{min}(G,k) + k - 2$. □

3 The Average-Set-Eccentricity

In this section we consider the k-set-eccentricity resulting from defining the distance between two sets as the average distance between a vertex from each.

For $k \geq 2$, we denote the ave-k-set-radius of G as

$$rad_{ave}(G,k) = \min\{\, e_{ave}(P) : P \subseteq V,\ |P| = k-1 \,\},$$

and the ave-k-set-diameter of G as

$$diam_{ave}(G,k) = \max\{\, e_{ave}(P) : P \subseteq V,\ |P| = k-1 \,\}.$$

We start with results which relate the ordinary radius and the ave-k-set-radius. The first bound might not be suprising, but the fact that $k=4$ is an exception does seem suprising.

Theorem 8 *Let G be a connected graph, and let $k \geq 2$ an integer. Then*

$$rad\ G \leq diam\ G \leq 2\, rad_{ave}(G,k),$$

and this bound is sharp for $k \neq 4$.

Proof. Let S be a set of $k-1$ vertices such that $e_{ave}(S) = rad_{ave}(G,k)$. Let v and w be any vertices in G such that $d(v,w) = diam\ G$. Then

$$\begin{aligned}
rad_{ave}(G,k) &= e_{ave}(S) \\
&\geq \frac{1}{2}\left(d_{ave}(v,S) + d_{ave}(w,S)\right) \\
&= \frac{1}{2(k-1)}\left(\sum_{x\in S} d(v,x) + \sum_{x\in S} d(w,x)\right) \\
&\geq \frac{1}{2(k-1)}\left(\sum_{x\in S} d(v,w)\right) \\
&= \frac{1}{2} d(v,w) \\
&= \frac{1}{2}\, diam\ G.
\end{aligned}$$

That this bound is sharp for k odd may be seen as follows. Let $G \cong C_{2r}$ be a cycle on $2r$ vertices. Then $rad_{ave}(G,k) = r/2$, achieved by taking $(k-1)/2$ distinct pairs of antipodal vertices. On the other hand, $rad\ G = r$.

That the bound is also sharp for $k \geq 6$ even, may be seen as follows.

Let $s = k-2 \geq 4$ be an even integer. We construct a graph G with average $(s+2)$-radius s and normal radius $2s$. By the above bound, it suffices to show that G has average $(s+2)$-radius at most s and radius at least $2s$.

We define G in two steps. We define first a gadget H which has a large number of vertices, is connected, and which has s designated vertices, called gates. The gadget is defined by first specifying for each vertex a vector giving the distance to each gate, and then adding edges to achieve this. Then we take $s+1$ new vertices, called middles, and $s+1$ copies of the gadget, and identify the gates with the middles in every possible way. The exact details are explained below.

First we need to define the gadget. We define a sequence of s multisets $w_0, \ldots, w_{s-1}$ each consisting of s nonnegative integers. We will think of the multisets as ordered in nondecreasing order.

First we set $w_0 = \{0, s+1, \ldots, s+1\}$. Then the ordered multiset w_{i+1} is obtained by incrementing and decrementing some of the entries in w_i. (By increment and decrement we mean increase and decrease by 1.) For $0 \leq i < s/2 - 1$, to get from w_i to w_{i+1}, increment the first entry, decrement the next $i+2$ entries, and increment the final i entries. To get from $w_{s/2-1}$ to $w_{s/2}$, increment the first entry, then decrement the second and the final entries. For $s/2 \leq i < s-1$, to get from w_i to w_{i+1}, increment the first entry, increment the next $s-i-2$ entries, and decrement the final $s-i$ entries.

For example, for $s = 4$ the ordered multisets are 0555, 1445, 2344 and 3333. For $s = 6$ they are 077777, 166777, 255678, 345677, 455566 and 555555. For $s = 10$, for example, the changes can be depicted as illustrated below. This picture also shows that the process of getting from w_i to w_{i+1} is in some sense the inverse of the process of getting from w_{s-i-1} to w_{s-i} if one ignores the first entry.

	0	11	11	11	11	11	11	11	11	11
$w_1 - w_0$	$+$	$-$	$-$							
$w_2 - w_1$	$+$	$-$	$-$	$-$						$+$
$w_3 - w_2$	$+$	$-$	$-$	$-$	$-$				$+$	$+$
$w_4 - w_3$	$+$	$-$	$-$	$-$	$-$	$-$		$+$	$+$	$+$
$w_5 - w_4$	$+$	$-$								$-$
$w_6 - w_5$	$+$	$+$	$+$	$+$		$-$	$-$	$-$	$-$	$-$
$w_7 - w_6$	$+$	$+$	$+$				$-$	$-$	$-$	$-$
$w_8 - w_7$	$+$	$+$						$-$	$-$	$-$
$w_9 - w_8$	$+$								$-$	$-$
	9	9	9	9	9	9	9	9	9	9

Let us note some properties. It can be checked that the multiset w_i remains ordered in nondecreasing order throughout. The sum of the entries of w_i decreases by 1 each time: it's $s^2 - 1 - i$.

Now we are in a position to describe the gadget. For each multiset w_i and every possible ordering of the elements of w_i, create a vector with that ordering. We denote the vector $(v_1, v_2, \ldots, v_s)$ by (v_i). Then for each vector create a vertex. Join two vertices iff their associated vectors differ by at most 1 in each component. For example, if $s = 4$ there will be vertices 4154 and 4234 and 5144 inter alia, and 5144 is joined to both 4154 and 4234, but 4154 and 4234 are not adjacent.

The s vertices arising from w_0 we call the gates. For $1 \leq i \leq s$, let g_i be the gate whose vector has 0 in the ith position.

We will next prove:

Claim 9 *If a vertex v has vector (v_i), then the distance from v to g_i is v_i.*

Proof. That the distance is at least v_i is immediate from the way edges are added: traversing an edge can change the ith component by at most 1.

To see that this value is obtainable, we argue by induction on v_i. The result is trivial if $v_i = 0$ as then $v = g_i$.

If the value v_i is the minimum of the vector, then consider the vertex resulting from the same ordering of $w_{v_i - 1}$ as gave rise to v from w_{v_i}. This vertex is adjacent to v and closer to g_i.

If the value v_i is the second smallest, suppose that a is the smallest value. If $v_i = a + 1$ then suppose j is the index such that $v_j = a$: decrementing v_i and incrementing v_j yields a valid vector. This vertex is adjacent to v and closer to g_i. If $v_i > a + 1$, then consider the vertex resulting from the same ordering of w_{a+1} as gave v from w_a. This vertex is adjacent to v and closer to g_i.

If the value v_i is neither the minimum nor the second smallest, then we note a property of the w_i. Apart from the first element of w_i, two consecutive values of w_i differ by at most 1.

So there is a j such that $v_j = v_i - 1$. If we take the vector for v and decrement the ith entry while incrementing the jth entry we obtain a new valid vector. This vertex is adjacent to v and closer to g_i.

For example, in the case $s = 4$, the vertex corresponding to 4154 has distance 5 to g_3 by the moves 5144, 4234, 4324, 5414, 5505. □

Now we construct the graph G as described above: We take a set M of $s + 1$ vertices, called middles. We take $s + 1$ copies of the gadget, and identify the s gates of each gadget with s middles so that each gadget's gates are a different subset of M.

Claim 10 *For any vertex v in G, the total distance from v to M is $s(s + 1)$.*

Proof. Suppose $v \in M$. The distance between any pair of middles is $s + 1$. So the conclusion holds for v.

Suppose $v \notin M$. There are s middles it can reach directly, and for the final one it must go via one of the other middles. Therefore, v's distance to the final one is $s + 1$ plus the distance from v to the nearest gate. By Claim 9, v's distance to the s gates in its gadget is the sum of the entries of the corresponding multiset w_i. It is easily checked that for each i, the sum of the entries of the multiset w_i with the first entry repeated is $s^2 - 1$. Since $(s + 1) + (s^2 - 1) = s(s + 1)$, we are done. □

Claim 11 *The eccentricity of every vertex v in G is at least $2s$.*

Proof. This uses the symmetric nature of the construction of the w_i. Consider any vertex v of some gadget H. Without loss of generality, the vector (v_i) for v is ordered in increasing order; and the gate g_i in H is identified with the middle vertex m_i for $1 \leq i \leq s$. By Claim 9, v is at distance v_i in H from m_i for $1 \leq i \leq s$. In particular, v is at distance v_1 from the nearest gate m_1. Now consider the gadget H' which does not have m_1 as a gate. Without loss of generality, the gates in H' are ordered $m_2, m_3, \ldots, m_{s+1}$. Then consider the vertex z in H' with vector

$$(2s - v_2, 2s - v_3, \ldots, 2s - v_s, s - 1 - v_1)$$

This vector arises from w_{s-1-v_1}, as can be checked.

By Claim 9, z is at distance $2s - v_i$ in H' from m_i for $2 \leq i \leq s$ and at distance $s - 1 - v_1$ from m_{s+1}. Thus, $d(v, m_i) + d(m_i, z) \geq 2s$ for $2 \leq i \leq s$. This means that if the path from v to z goes via m_i, $2 \leq i \leq s$, then it has length at least $2s$. If none of these middles is on the path, then the path goes from v to m_1 to m_{s+1} to z. But $d(v, m_1) + d(m_1, m_{s+1}) + d(m_{s+1}, z) = (s - 1) + (s + 1) = 2s$. Thus the distance from v to z is at least $2s$. This completes the proof of the claim and of Theorem 8. □

The next result provides the optimal bound for $k = 4$.

Theorem 12 *Let G be a connected graph. Then*

$$rad\ G \leq \frac{9}{5} rad_{ave}(G, 4),$$

and this bound is sharp.

Proof. Set $rad\, G = r$. Let $C = \{c_1, c_2, c_3\}$ be a subset of $V(G)$ for which $e_{ave}(C) = rad_{ave}(G, 4)$. For $\ell = 1, 2, 3$, let v_ℓ be an eccentric vertex of c_ℓ, and so $d(v_\ell, c_\ell) \geq r$.

Then we can bound the average distance from v_1 to C in two ways by the triangle inequality:

$$\begin{aligned} 3d_{ave}(v_1, C) &= d(v_1, c_1) + [d(v_1, c_2) + d(v_1, c_3)] \\ &\geq r + d(c_2, c_3), \end{aligned}$$

and

$$\begin{aligned} 3d_{ave}(v_1, C) &= d(v_1, c_1) + [d(v_1, c_2)] + [d(v_1, c_3)] \\ &\geq r + [r - d(c_1, c_2)] + [r - d(c_1, c_3)] \\ &= 3r - d(c_1, c_2) - d(c_1, c_3). \end{aligned}$$

If we establish similar bounds for $d_{ave}(v_2, C)$ and $d_{ave}(v_3, C)$, and sum, we obtain the two inequalities:

$$3 \sum_\ell d_{ave}(v_\ell, C) \geq 3r + \sum_{i<j} d(c_i, c_j),$$

and

$$3 \sum_\ell d_{ave}(v_\ell, C) \geq 9r - 2 \sum_{i<j} d(c_i, c_j).$$

Double the first inequality added to the second inequality gives $9 \sum_\ell d_{ave}(v_\ell, C) \geq 15r$, so that

$$e_{ave}(C) \geq \max_\ell d_{ave}(v_\ell, C) \geq 5r/9.$$

That this bound is sharp is seen by considering a cycle whose order is a multiple of 6. The ave-4-set-eccentricity is achieved by placing three vertices equally spaced on the cycle. □

At the other end of the scale, the ave-k-set-radius cannot be much more than the ordinary radius. We shall denote the set of all end-vertices of a tree T by $W(T)$.

Theorem 13 *Let G be a connected graph, and let $k \geq 2$ be an integer. Then*

$$rad_{ave}(G, k) \leq rad\, G + \frac{k^2}{8(k-1)},$$

and this bound is sharp.

Proof. It suffices to prove the inequality for a tree $T = (V, E)$, as a graph G always has a spanning tree of the same radius and the ave-k-set-radius of this spanning tree is at least that of G.

The proof for a fixed k is by induction on the order n of the tree.

Case 1. $n = k$.

We prove the stronger result that $diam_{ave}(T,k) \le rad\,T + k^2/(8(k-1))$. Let $v \in V$ and let $S \subset V$ be any $(n-1)$-set. If $e(v) = \ell$, then

$$d_{ave}(v,S) \le \frac{1}{n-1}\Big(1+2+\ldots+(\ell-1)+(n-\ell)\ell\Big) = \ell - \frac{\ell^2-\ell}{2(n-1)}.$$

As this bound is increasing in ℓ, we have, since $\ell \le diam\,T \le 2\,rad\,T$,

$$d_{ave}(v,S) \le 2\,rad\,T - \frac{2(rad\,T)^2 - rad\,T}{n-1};$$

or, equivalently,

$$d_{ave}(v,S) - rad\,T \le rad\,T - \frac{2(rad\,T)^2 - rad\,T}{n-1}.$$

The right-hand side of the last inequality achieves its maximum for $rad\,T = n/4$. Hence

$$d_{ave}(v,S) - rad\,T \le \frac{n}{4} - \frac{n^2-2n}{8(n-1)} = \frac{k^2}{8(k-1)},$$

as desired.

Case 2. $n > k$ and $k-1 \ge n - |W(T)|$.

Let $S \subset V$ be any $(k-1)$-set containing every non end-vertex of T. Every vertex in $V-S$ is an end-vertex of T and is adjacent to a vertex of S. For a vertex $v \in V-S$, let T_v be the tree induced by $S \cup \{v\}$. For $v \in S$, choose any vertex $u \in V-S$ and let T_v be the tree induced by $S \cup \{u\}$. Then T_v contains k vertices and the result of Case 1, applied to T_v, yields

$$d_{ave}(v,S) \le diam_{ave}(T_v,k) \le rad\,T_v + \frac{k^2}{8(k-1)} \le rad\,T + \frac{k^2}{8(k-1)}.$$

Case 3. $n > k$ and $k-1 < n - |W(T)|$.

Let $T' = T - W(T)$. Then $rad\,T' = rad\,T - 1$. Applying the inductive hypothesis to T' we have

$$rad_{ave}(T',k) \le rad\,T' + \frac{k^2}{8(k-1)} = rad\,T - 1 + \frac{k^2}{8(k-1)}.$$

Let S be a $(k-1)$-set of vertices in T' satisfying $e_{ave}(S) = rad_{ave}(T',k)$. If $v \in V(T')$, then $d_{ave}(v,S) \le rad\,T - 1 + k^2/(8k-8)$. On the other hand, if $v \in W(T)$, then v is an adjacent to a vertex v' of T', and so $d_{ave}(v,S) = d_{ave}(v',S) + 1 \le rad\,T + k^2/(8(k-1))$, as desired.

That the bound is sharp may be seen by considering the broom $T(k/2,1,k/2)$ of order k and diameter $k/2$, if k is even. □

We remark that if G is a connected graph of sufficiently large radius, and $k \ge 3$, then the upper bound of Theorem 13 can probably be improved to $rad_{ave}(G,k) \le rad\ G + \frac{k}{16} + O(1)$. This bound is attained by the four-legged octopus $T(4,r,r)$.

Theorem 14 *Let G be a connected graph of order n, and let $k \ge 2$ be an integer. Then*

$$rad_{ave}(G,k) \le \left\lfloor \frac{n}{2} \right\rfloor,$$

and this bound is sharp.

Proof. The proof for fixed k is by induction on n. It suffices to prove the claim for a tree $T = (V, E)$. We distinguish two cases.

Case 1. $k - 1 \geq n - |W(T)|$.

Let $S \subset V$ be any $(k-1)$-set containing every non-end-vertex of T. Since every vertex of T is in S or adjacent to a vertex of S, we have

$$d_{ave}(v, S) \leq \frac{1}{k-1}\Big(1 + 2 + \ldots + (k-1)\Big) = \frac{k}{2} \leq \left\lfloor \frac{n}{2} \right\rfloor,$$

for every $v \in V$. Hence $rad_{ave}(T, k) \leq \lfloor n/2 \rfloor$.

Case 2. $k - 1 < n - |W(T)|$.

Let $T' = T - W(T)$. Then T' is a tree of order $n' = n - |W(T)| \geq k$. By the inductive hypothesis,

$$rad_{ave}(T', k) \leq \left\lfloor \frac{n'}{2} \right\rfloor \leq \left\lfloor \frac{n}{2} \right\rfloor - 1.$$

Let S be a $(k-1)$-set of vertices in T' satisfying $e_{ave}(S) = rad_{ave}(T', k)$. If $v \in V(T')$, then $d_{ave}(v, S) \leq \lfloor n/2 \rfloor - 1$. On the other hand, if $v \in W(T)$, then v is adjacent to a vertex v' of T', and so

$$d_{ave}(v, S) \leq d_{ave}(v', S) + 1 \leq \lfloor n/2 \rfloor.$$

Hence, $rad_{ave}(T, k) \leq \min_v d_{ave}(v, S) \leq \lfloor n/2 \rfloor$.

To see that bound is sharp, take G to be a path and $k = 2$. $\Box$

Remark: To find $rad_{ave}(P_n, k)$ for a path, notice that unless k and n are both even, any symmetric placement of k vertices on the path is optimal. If k and n are both even, then one (not unique) optimal placement is to use precisely one of the central vertices and the remaining $k-2$ vertices placed symmetrically.

The ordinary diameter and the k-set-diameter of a graph are related as follows.

Theorem 15 *Let $G = (V, E)$ be a connected graph. Then*

$$diam_{ave}(G, k) \leq diam\ G \leq diam_{ave}(G, k) + \frac{k-1}{2},$$

and these bounds are sharp.

Proof. Let S be a $(k-1)$-element subset of vertices of G. Then for every vertex v of G, v is at distance at most *diam* G from each vertex of S, so $d_{ave}(v, S) \leq diam\ G$. Thus, $e_{ave}(S) \leq diam\ G$, whence $diam_{ave}(G, k) \leq diam\ G$. This establishes the lower bound. That this bound is sharp, may be seen by considering the octopus $G \cong T(k, r, r)$. Letting S consist of $k-1$ end-vertices of G, and let v denote the end-vertex of G that does not belong to S. Then $d_{ave}(v, S) = 2r = diam\ G$, whence $diam_{ave}(G, k) \geq diam\ G$. Consequently, $diam_{ave}(G, k) = diam\ G$.

Next we consider the upper bound. Let u and v be two vertices of G at maximum distance apart, and let P be a shortest u–v path. Then, letting S be the first $k-1$ vertices on P, we have $d_{ave}(v,S) = diam\ G - (k-1)/2$. Thus, $diam_{ave}(G,k) \geq e_{ave}(S) \geq d_{ave}(v,S) = diam\ G - (k-1)/2$. That this bound is sharp may be seen by taking G to be a path on at least k vertices. □

Theorem 16 *Let G be a connected graph, and let $k \geq 2$ be an integer. Then*

$$diam_{ave}(G,k) \leq n - k/2,$$

and this bound is sharp.

Proof. It is not hard to show that the total distance to the j vertices farthest away is at most $jn - \binom{j+1}{2}$ attained only if the graph is a path. □

The k-set-radius and k-set-diameter of a graph are related by the following inequality.

Theorem 17 *Let $G = (V,E)$ be a connected graph, and let $k \geq 2$ be an integer. Then*

$$diam_{ave}(G,k) \leq 2\, rad_{ave}(G,k).$$

and this bound is sharp.

Proof. By Theorems 8 and 15, $diam_{ave}(G,k) \leq diam\ G \leq 2\, rad_{ave}(G,k)$.

To see that this bound is sharp, let G denote the lexicographic product of an odd path with the empty graph on k vertices: that is, take disjoint sets $V_1, V_2, \ldots, V_{2r+1}$ each with k vertices, and for $i = 1, \ldots, 2r$ make all the vertices in V_i adjacent to all the vertices in V_{i+1}. Then $diam_{ave}(G,k) = diam\ G = 2r$ and $rad_{ave}(G,k) = rad\ G = r$. □

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